Two-Seller Competition: Oblivious equilibrium

Progress note — transcription of handwritten notes

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In this note we present four different models of competition where the sellers don't observe the full state. We restrict this note to the forgetting case.

1 Elements of the model

- Two symmetric providers, indexed by $j \in \{A, B\}$.
- Two products, indexed by $i \in \{1, 2\}$.
- Success probabilities: $0 < p_1 < p_2 < 1$.
- Costs: $c_1 < c_2$.
- Revenue: $R > c_2$.

2 Customer beliefs.

Recall that in the forgetting model, past observations are discounted by a factor ρ whenever a new outcome is observed. Thus, after k interactions with the seller, the total effective information satisfies

$$S_k + F_k = \rho(S_{k-1} + F_{k-1}) + 1,$$

with $S_0 + F_0 = 0$. Iterating this expression yields

$$S_k + F_k = 1 + \rho + \cdots + \rho^{k-1} = \frac{1 - \rho^k}{1 - \rho} \quad \to 1/(1 - \rho) \quad \text{as } k \to \infty.$$

To study the limiting behavior, we normalize the state as

$$x_s = (1 - \rho)S_k$$

$$x_f = (1 - \rho)F_k$$

Since $x_s + x_f = 1 - \rho^k \to 1$, we focus on the limiting boundary $x_s + x_f = 1$, which can be represented by a single variable $x = x_s$, with $x_f = 1 - x$. Then $x \in [0, 1]$ when $\rho \in [1/2, 1)$.

For each seller we will call this normalized state variable x_j for $j \in \{A, B\}$ that will represent the proportion of successes for provider j in the long run. The customer observes the state

$$x = (x_A, x_B),$$

Her beliefs are

$$\theta_A \mid x_A \sim \text{Beta}(1 + \frac{x_A}{1 - \rho}, 1 + \frac{1 - x_A}{1 - \rho}), \tag{1}$$

$$\theta_B \mid x_B \sim \text{Beta}(1 + \frac{x_B}{1 - \rho}, 1 + \frac{1 - x_B}{1 - \rho}). \tag{2}$$

Customer choice. The customer uses Thompson sampling (TS), drawing independently from the two belief distributions and choosing

$$y_t = \begin{cases} A, & \widetilde{\theta}_A > \widetilde{\theta}_B, \\ B, & \widetilde{\theta}_B > \widetilde{\theta}_A. \end{cases} \tag{3}$$

Thus the selection probabilities are

$$D_A(x_A, x_B) = \mathbb{P}(\widetilde{\theta}_A > \widetilde{\theta}_B \mid x_A, x_B), \qquad D_B(x_A, x_B) = 1 - D_A(x_A, x_B). \tag{4}$$

3 Timing, state transitions, and strategies

Timing.

1. The customer observes $x = (x_A, x_B)$, while each seller j observes only its own reputation x_j.
2. The customer chooses a seller using Thompson sampling.
3. The selected seller chooses a product $i \in \{1, 2\}$ according to its strategy $\sigma_j(\cdot \mid x_j)$ and receives profit $R - c_i$.
4. The customer observes the outcome, but not the product.
5. The selected seller's reputation is updated, while the other seller's reputation remains unchanged.

State transitions. Superscripts s and f denote success and failure, respectively. The state transitions are

$$\begin{aligned} T_A^s(x) &= (\rho x_A + 1 - \rho, x_B), \\ T_A^f(x) &= (\rho x_A, x_B), \\ T_B^s(x) &= (x_A, \rho x_B + 1 - \rho), \\ T_B^f(x) &= (x_A, \rho x_B). \end{aligned} \tag{5}$$

Strategies. We consider stationary mixed strategies that depend only on the seller's own reputation. For each seller $j \in \{A, B\}$, a strategy is a mapping

$$\sigma_j : [0, 1] \to \Delta(\{1, 2\}), \tag{6}$$

where $\sigma_j(i \mid x_j)$ denotes the probability that seller j chooses product i when selected at reputation x_j.

4 Oblivious equilibrium: Distribution

Oblivious approximation: Suppose that each seller $j \in \{A, B\}$ assumes that its competitor's reputation follows a stationary distribution μ_{-j} on $[0, 1]$.

Instead of observing the competitor's current reputation, seller j computes its expected demand by averaging over this distribution:

$$d_j(x_j; \mu_{-j}) = \int_0^1 D(x_j, y)\, \mu_{-j}(dy). \tag{7}$$

Thus, each seller's demand depends only on its own reputation and the conjectured distribution of its competitor. When optimizing, the seller treats this distribution as fixed.

Seller's problem: Let $U_j(x_j)$ denote seller j's optimal expected discounted profit before customer selection. Given the conjectured distribution μ_{-j}, the Bellman equation is

$$\begin{aligned} U_j(x_j) = \gamma[1 - d_j(x_j;\mu_{-j})]U_j(x_j) \\ + d_j(x_j;\mu_{-j}) \max_{i\in\{1,2\}} \Big\{R - c_i \\ + \gamma\left[p_i U_j(T_\rho^s(x_j)) + (1-p_i)U_j(T_\rho^f(x_j))\right]\Big\}. \end{aligned} \tag{8}$$

If the seller is not selected, it receives no profit and its reputation remains unchanged. If selected, it chooses a product to maximize its current profit and expected future value.

Stationary distribution: Given a strategy σ_j and a conjectured distribution μ_{-j}, the probability of success conditional on seller j being selected is

$$p_{\sigma_j}(x_j) = \sigma_j(1 \mid x_j)p_1 + \sigma_j(2 \mid x_j)p_2. \tag{9}$$

Given a strategy profile $\sigma = (\sigma_A, \sigma_B)$, let $P^\sigma(x, C)$ denote the probability that the joint reputation state belongs to a measurable set $C \subseteq [0,1]^2$ in the next period, given the current state $x = (x_A, x_B)$. The transition probabilities are

$$\begin{aligned} P^\sigma(x,C) = D(x_A,x_B)\Big[p_{\sigma_A}(x_A)\mathbf{1}\{T_A^s(x)\in C\} + (1-p_{\sigma_A}(x_A))\mathbf{1}\{T_A^f(x)\in C\}\Big] \\ + [1-D(x_A,x_B)]\Big[p_{\sigma_B}(x_B)\mathbf{1}\{T_B^s(x)\in C\} + (1-p_{\sigma_B}(x_B))\mathbf{1}\{T_B^f(x)\in C\}\Big] \end{aligned} \tag{10}$$

Oblivious equilibrium: An oblivious equilibrium is a mixed strategy σ^*, a reputation distribution μ^*, and a joint stationary distribution ν^* satisfying the following conditions.

Optimality:

$$\begin{aligned} \operatorname{supp}\sigma^*(\cdot \mid x) \subseteq \arg\max_{i\in\{1,2\}} \Big\{R - c_i + \gamma\big[p_i U_{\mu^*}(T_\rho^s(x)) \\ + (1-p_i)U_{\mu^*}(T_\rho^f(x))\big]\Big\}. \end{aligned} \tag{11}$$

Consistency:

$$\begin{aligned} \nu^* &= \nu^* P^{\sigma^*}, \\ \mu^*(C) &= \nu^*(C\times[0,1]) = \nu^*([0,1]\times C), \qquad \forall C \subseteq [0,1] \text{ measurable.} \end{aligned} \tag{12}$$

Here P^{σ^*} denotes the joint transition kernel of the actual two-seller game when both sellers follow σ^*.

The first condition requires each seller to choose only optimal products given the conjectured distribution μ^*. The second requires this distribution to coincide with the marginal distributions generated when both sellers follow σ^* in the actual game.

5 Oblivious equilibrium: Fixed quality

Seller's conjecture: Suppose that each seller $j \in \{A, B\}$ represents its competitor by a fixed probability $p_{0,j} \in (0,1)$, corresponding to the long-run average of the competitor's Thompson sampling draws.

Instead of considering the competitor's reputation distribution, seller j computes its expected demand as

$$d(x_j; p_{0,j}) = \mathbb{P}(\widetilde{\theta}_j > p_{0,j} \mid x_j). \tag{13}$$

Thus, each seller faces the original problem with an outside option of fixed quality $p_{0,j}$. When optimizing, the seller treats this probability as fixed.

Seller's problem: Let $U_{p_{0,j}}(x_j)$ denote seller j's optimal expected discounted profit before customer selection, given $p_{0,j}$.

The Bellman equation is

$$\begin{aligned} U_{p_{0,j}}(x_j) = \gamma[1 - d(x_j; p_{0,j})]U_{p_{0,j}}(x_j) \\ + d(x_j; p_{0,j}) \max_{i\in\{1,2\}} \Big\{ R - c_i \\ + \gamma \left[p_i U_{p_{0,j}}(T_\rho^s(x_j)) + (1-p_i) U_{p_{0,j}}(T_\rho^f(x_j)) \right] \Big\}. \end{aligned} \tag{14}$$

Oblivious equilibrium: We consider a symmetric equilibrium in which both sellers use the same mixed strategy σ^* and the same conjectured probability p_0^*. An oblivious equilibrium is a triple (σ^*, p_0^*, ν^*) satisfying the following conditions.

Optimality.

$$\begin{aligned} \operatorname{supp} \sigma^*(\cdot \mid x) \subseteq \arg\max_{i\in\{1,2\}} \Big\{ R - c_i + \gamma \big[p_i U_{p_0^*}(T_\rho^s(x)) \\ + (1-p_i) U_{p_0^*}(T_\rho^f(x)) \big] \Big\}. \end{aligned} \tag{15}$$

Consistency.

$$\begin{aligned} \nu^* &= \nu^* P^{\sigma^*}, \\ p_0^* &= \int_{[0,1]^2} \frac{1-\rho+x_A}{3-2\rho}\, \nu^*(dx_A, dx_B) \\ &= \int_{[0,1]^2} \frac{1-\rho+x_B}{3-2\rho}\, \nu^*(dx_A, dx_B). \end{aligned} \tag{16}$$

Here P^{σ^*} denotes the joint transition kernel of the actual two-seller game when both sellers follow σ^*.

The optimality condition requires each seller to assign positive probability only to products that maximize its expected discounted profit given p_0^*. The consistency condition requires the joint reputation distribution to be stationary and p_0^* to coincide with the long-run average of the competitor's Thompson sampling draws.

6 Oblivious equilibrium: Average success probability

Seller's conjecture: Suppose that each seller $j \in \{A, B\}$ represents its competitor by a fixed probability $p_{0,j} \in [p_1, p_2]$, corresponding to the competitor's long-run success rate among its sales.

Instead of considering the competitor's reputation distribution, seller j computes its expected demand as

$$d(x_j; p_{0,j}) = \mathbb{P}(\widetilde{\theta}_j > p_{0,j} \mid x_j). \tag{17}$$

Thus, each seller faces the original forgetting monopoly problem with an outside option of fixed quality $p_{0,j}$. When optimizing, the seller treats this probability as fixed. Unlike the previous model, $p_{0,j}$ represents the competitor's average probability of success among its sales, rather than the average of its Thompson sampling draws.

Seller's problem: Let $U_{p_{0,j}}(x_j)$ denote seller j's optimal expected discounted profit before customer selection, given $p_{0,j}$.

The Bellman equation is

$$\begin{aligned} U_{p_{0,j}}(x_j) = \gamma[1 - d(x_j; p_{0,j})]U_{p_{0,j}}(x_j) \\ + d(x_j; p_{0,j}) \max_{i\in\{1,2\}} \Big\{ R - c_i \\ + \gamma \left[p_i U_{p_{0,j}}(T^s_\rho(x_j)) + (1-p_i) U_{p_{0,j}}(T^f_\rho(x_j)) \right] \Big\}. \end{aligned} \tag{18}$$

The seller solves the same problem as in the fixed-quality model. The difference is the consistency condition used to determine $p_{0,j}$.

Oblivious equilibrium: We consider a symmetric equilibrium in which both sellers use the same mixed strategy σ^* and the same conjectured probability p_0^*. An oblivious equilibrium is a triple (σ^*, p_0^*, ν^*) satisfying the following conditions.

Optimality.

$$\begin{aligned} \operatorname{supp} \sigma^*(\cdot \mid x) \subseteq \arg\max_{i\in\{1,2\}} \Big\{ R - c_i + \gamma \big[p_i U_{p_0^*}(T^s_\rho(x)) \\ + (1-p_i) U_{p_0^*}(T^f_\rho(x)) \big] \Big\}. \end{aligned} \tag{19}$$

Consistency.

$$\begin{aligned} \nu^* &= \nu^* P^{\sigma^*}, \\ p_0^* &= \frac{\displaystyle\int_{[0,1]^2} D_B(x_A, x_B)\, p_{\sigma^*}(x_B)\, \nu^*(dx_A, dx_B)}{\displaystyle\int_{[0,1]^2} D_B(x_A, x_B)\, \nu^*(dx_A, dx_B)} \\ &= \frac{\displaystyle\int_{[0,1]^2} D_A(x_A, x_B)\, p_{\sigma^*}(x_A)\, \nu^*(dx_A, dx_B)}{\displaystyle\int_{[0,1]^2} D_A(x_A, x_B)\, \nu^*(dx_A, dx_B)}. \end{aligned} \tag{20}$$

Here P^{σ^*} denotes the joint transition kernel of the actual two-seller game when both sellers follow σ^*, and $p_{\sigma^*}(x)$ is the probability of success conditional on being selected at reputation x.

The optimality condition requires each seller to assign positive probability only to products that maximize its expected discounted profit given p_0^*. The consistency condition requires the joint reputation distribution to be stationary and p_0^* to coincide with the competitor's success rate among its sales in the actual game. Weighting by the selection probability and dividing by the seller's purchase frequency makes this an average among sales. The probability p_0^* is generated by the strategy and does not represent an intrinsic quality of the seller.

7 Oblivious equilibrium: Conditional distribution

Seller's conjecture: Suppose that each seller $j \in \{A, B\}$ assumes that its competitor's reputation follows a conditional distribution $\kappa_j(\cdot \mid x_j)$ on $[0,1]$, indexed by the seller's own reputation x_j before customer selection.

Seller j computes its expected demand as

$$d_j(x_j; \kappa_j) = \int_0^1 D(x_j, y)\, \kappa_j(dy \mid x_j). \tag{21}$$

The seller uses its own reputation as information about its competitor, without observing the competitor's current state. When optimizing, it holds the entire kernel κ_j fixed, using its prescribed conditional law at each future own state.

Seller's problem: Let $U_{\kappa_j}(x_j)$ denote seller j's optimal expected discounted profit before customer selection, given κ_j. The Bellman equation is

$$\begin{aligned} U_{\kappa_j}(x_j) = \gamma[1 - d_j(x_j; \kappa_j)] U_{\kappa_j}(x_j) \\ + d_j(x_j; \kappa_j) \max_{i \in \{1,2\}} \Big\{ R - c_i \\ + \gamma \left[p_i U_{\kappa_j}(T_\rho^s(x_j)) + (1 - p_i) U_{\kappa_j}(T_\rho^f(x_j)) \right] \Big\}. \end{aligned} \tag{22}$$

Oblivious equilibrium: We consider a symmetric equilibrium in which both sellers use the same mixed strategy σ^* and the same conjectured conditional kernel κ^*. A conditional oblivious equilibrium is a triple $(\sigma^*, \kappa^*, \nu^*)$, where ν^* is a symmetric joint reputation distribution with common marginal μ^*, satisfying the following conditions.

Optimality.

$$\begin{aligned} \operatorname{supp} \sigma^*(\cdot \mid x) \subseteq \arg \max_{i \in \{1,2\}} \Big\{ R - c_i + \gamma \big[p_i U_{\kappa^*}(T_\rho^s(x)) \\ + (1 - p_i) U_{\kappa^*}(T_\rho^f(x)) \big] \Big\}. \end{aligned} \tag{23}$$

Consistency.

$$\begin{aligned} \nu^* &= \nu^* P^{\sigma^*}, \\ \mu^*(C) &= \nu^*(C \times [0,1]) = \nu^*([0,1] \times C), \\ \nu^*(C \times E) &= \int_C \kappa^*(E \mid x)\, \mu^*(dx), \qquad C, E \subseteq [0,1] \text{ measurable.} \end{aligned} \tag{24}$$

Here P^{σ^*} denotes the joint transition kernel of the actual two-seller game when both sellers follow σ^*. The last condition requires the conjectured kernel to coincide with the conditional distribution generated by this joint stationary law, μ^*-almost everywhere.

On a finite reputation grid, this means $\kappa^*(y \mid x) = \nu^*(x, y)/\mu^*(x)$ whenever $\mu^*(x) > 0$. At zero-mass grid states, we complete the conjecture using the unconditional marginal μ^* and require optimality under that completion.

The optimality condition requires each seller to choose only optimal products given κ^*. Unlike the distribution model, consistency is imposed on the conditional law for μ^*-almost every own state, rather than only on the unconditional marginal. The policy still depends only on own reputation; the seller does not observe the full joint state or track a belief based on the entire history of play.